4601 Fairfax Drive N, Suite 600 | Arlington, VA 22203
+1 833 358 3623

fluenceenergy.com

Docket No. RM26-4-000
Filed: November 21, 2025

Debbie-Anne A. Reese
Secretary
Federal Energy Regulatory Commission
888 First Street, NE, Room 1A
Washington, DC 20426

Submitted by:
Suzanne Leta
Vice President, Policy and Advocacy
Fluence
suzanne.leta@fluenceenergy.com

RE: Comments on Advance Notice of Proposed Rulemaking Regarding the *Interconnection of Large Loads Pursuant to the Secretary's Authority Under Section 403 of the Department of Energy Organization Act*

Fluence appreciates the opportunity to comment on the Advance Notice of Proposed Rulemaking (ANOPR) and is supportive of the Federal Energy Regulatory Commission's ("FERC" or the "the Commission") efforts to ensure the timely and orderly interconnection of large loads to the transmission system.

Established in 2018, Fluence is an American company and global market leader in utility-scale energy storage products and services, as well as optimization software for storage. Fluence provides an ecosystem of offerings, including modular, scalable energy storage products, comprehensive service offerings, and AI-enabled optimization software. Projects with Fluence's integrated battery energy storage systems provide power to utilities, companies, and consumers across eight of the 10 U.S. ISO power markets, where 278 million Americans live, and the portfolio of over 80 energy storage projects utilizing our systems in the United States represents more than 1 million operating hours (or 41,667 days). The company is transforming the way we power our world by helping system operators, utilities, and power producers create more resilient and reliable electric grids.

Fluence is a leader in the onshoring of the grid-scale lithium iron phosphate (LFP) battery energy storage system (BESS) supply chain, and we pride ourselves on best-in-class quality, cybersecurity, and safety:

- Fluence can now deliver a domestic content system to customers with every major product and component of the system (including LFP cells) sourced from a U.S. manufacturing facility.

- Fluence actively manages its supply chain to prevent the use of controls hardware or software sourced from foreign adversary countries, reducing cybersecurity and regulatory risks.
- Fluence products are designed to meet or exceed relevant industry safety standards.

Utility-scale BESS has arrived and is already playing a key role in the electric grid. In 2017, the then-Energy Storage Alliance claimed that energy storage could reach 35 gigawatts (GW) of deployment by the end of 2025.[1] At the end of 2024, BESS had already reached an operating capacity of 26 GW[2] in the United States and was projected to add an additional 18.4 GW in 2025.[3] It is clear that BESS is proven and available, at scale.

Large loads like data centers often require capacity in the hundreds of megawatts to gigawatt scale and, importantly, require flexible operating characteristics that traditional generation cannot achieve on its own. BESS is ready to play a critical role in the all-of-the-above solution set that is required to meet this demand.

Below, please find comments from Fluence that further illustrate this key role.

FLUENCE COMMENTS ON ANOPR PRINCIPLES

Fluence supports the objectives of the ANOPR to standardize interconnection procedures for large loads and urges the Commission to adopt the following principles in any final action:

The definition of hybrid facilities should include energy storage. In the ANOPR, "hybrid facility" refers to large loads that intend to connect at the same point as new or existing generation resources. Since the Commission's procedures for large generator interconnection already recognize energy storage, the definition of "hybrid facility" for large loads should also explicitly include energy storage systems.

Curtailable or flexible loads should have an expedited pathway to interconnection. BESS is uniquely positioned to make large loads curtailable and flexible. BESS allows large loads to avoid drawing from the grid during peak demand periods by charging during off-peak periods

[1] Utility Dive. (n.d.). *Energy storage could hit 35 GW by 2025, new white paper says*. Retrieved from https://www.utilitydive.com

[2] U.S. Energy Information Administration. (2025, March 12). *U.S. battery capacity increased 66% in 2024*. Retrieved from https://www.eia.gov/todayinenergy/detail.php?id=64705

[3] U.S. Energy Information Administration. (2025, February 24). *Solar, battery storage to lead new U.S. generating capacity additions in 2025*. Retrieved from https://www.eia.gov/todayinenergy/detail.php?id=64586

and dispatching during times of peak demand, all while maintaining the characteristics of a dispatchable resource. This minimizes the system impact from interconnection and can minimize the need for expensive and time-consuming network upgrades. Given these highly desired attributes, the interconnection of hybrid facilities that include BESS should be expedited.

One example of an RTO/ISO that is easing pathways to interconnection is SPP through its Large Load Interconnection Tariff Revisions, which require High Impact Large Load Generator Assessment requests to be assessed outside the existing interconnection process.

Flexibility commitments should be met by large loads located adjacent to storage resources. The Commission should ensure that load commitments to flexibility can be met by resources physically and/or electrically adjacent to load.

Storage resources should be permitted to charge when they do not trigger the need for an upgrade. The Commission should strive to facilitate timely interconnection for hybrid facilities that do not require or at least defer network upgrades that delay interconnection. As such, the Commission should ensure any final action allows storage resources to charge will not trigger the need for upgrades. If unreasonable assumptions are applied to BESS that result in unnecessary and lengthy study processes, it will limit large loads from benefiting from the optionality that BESS can provide, which will ultimately exacerbate interconnection delays as opposed to quickening the pace of interconnection.

MARKET DESIGN PRINCIPLES TO SUPPORT BATTERY STORAGE DEPLOYMENT

All products should be biddable. BESS must have the ability to submit bids for all market products, including ramping and uncertainty products that are currently priced using loss of opportunity cost (LOO) calculations. The current LOO framework was designed for fuel-based generators with relatively static marginal costs and works adequately in that context. However, applying this same framework to energy storage creates fundamental market design flaws.

Unlike conventional generators where the opportunity cost can be calculated based on fuel costs and relatively straightforward dispatch alternatives, energy storage opportunity costs are inherently multi-interval and state dependent. The value of energy stored in a battery depends on future market clearing prices, forecast uncertainty, state of charge constraints, cycling limitations, and complex inter-temporal optimization across multiple settlement intervals and potentially multiple days. When an RTO/ISO forces energy storage to clear ramping or uncertainty products based on administratively determined LOO prices, rather than competitive bids, it systematically undervalues the flexibility these resources provide.

This forced clearing at sub-optimal prices creates several problems. First, it results in inefficient market outcomes where BESS resources may be committed to lower-value services when they could provide greater system benefit elsewhere. Second, it suppresses revenue streams that are essential for storage project economics, creating barriers to entry and reducing the incentive to deploy storage where it is most needed. Third, it prevents price formation that accurately reflects the true opportunity cost of the service being provided, potentially leading to inadequate compensation for flexibility that is increasingly critical to grid reliability.

The Commission should require all market products to be biddable by BESS resources, allowing market-based price formation to reveal the true value of the services being provided. This approach is consistent with Order 841's requirement for storage to participate in RTO/ISO markets on a comparable basis to other resources and with fundamental competitive market principles.

Multi-segment bids for all biddable products. BESS resources require multi-segment bidding capability for all market products, not just energy. While most RTOs/ISOs have implemented multi-segment bidding for energy products, few extend this capability to ancillary services, despite the fact that the economic rationale is even stronger for BESS in these markets.

For fuel-based generators, the marginal cost curve for providing ancillary services is relatively flat – providing 15 MW versus 20 MW of regulation has minimal impact on opportunity costs. However, for BESS, the opportunity cost can change dramatically with small changes in cleared capacity. Clearing an additional 5 MW of regulation may cross critical state-of-charge thresholds that materially impact the resource's ability to optimize across subsequent intervals, potentially foreclosing more valuable dispatch opportunities or requiring sub-optimal cycling.

Without multi-segment bidding, BESS resources face a binary choice. They can either bid at prices that reflect their maximum opportunity cost (resulting in frequent non-clearing and inefficient resource utilization) or bid at average opportunity costs (resulting in economic losses when the higher-cost segments clear). Neither outcome serves the market most efficiently.

Multi-segment bids would allow BESS resources to express their true supply curve, enabling economically efficient dispatch where the resource clears for the quantity where its marginal cost equals the market price. This would improve both resource economics and system efficiency by ensuring BESS is dispatched to the level where it provides the greatest net benefit.

The Commission should direct RTOs/ISOs to extend multi-segment bidding functionality to all products for which BESS resources are eligible to participate, recognizing that the state-dependent opportunity costs of BESS require more granular bidding capability than conventional resources.

BESS resources should have unrestricted ability to operate within battery parameters. RTOs/ISOs are, at times, imposing overly conservative operational restrictions on BESS resources that prevent these assets from providing their full technical capability to the grid. These restrictions take multiple forms, but all share the common characteristic of limiting storage operation below its actual technical parameters due to administrative constraints rather than physical limitations.

One example is CAISO's restriction preventing energy storage from carrying regulation capacity that exceeds the inverter capacity in a single direction. A 100 MW battery storage facility with bidirectional capability should be able to provide 200 MW of regulation capacity - 200 MW upward (discharge) when operating at the lower limit and 200 MW downward (charge) when operating at the upper limit. The inverter capacity constraint only applies to net power flow at any given instant, not to the capacity available for regulation over an operating range. By artificially restricting regulation capacity to equal the inverter rating, CAISO is leaving 50% of the BESS resource's regulation capability unutilized, harming both resource economics and system efficiency.

More broadly, RTOs/ISOs are implementing increasingly aggressive state-of-charge management algorithms that require large State of Charge (SOC) holdbacks to ensure resources can provide awarded capacity. While prudent SOC management has merit, current implementations are overly conservative and fail to account for the sophisticated optimization and forecasting capabilities that BESS operators employ. These holdback requirements effectively sterilize significant portions of the battery's capacity, reducing the resource's economic viability and the services available to the grid.

Adding to this challenge, each RTO/ISO has developed its own unique set of operational constraints, clearing algorithms, and SOC management approaches. This lack of standardization creates substantial barriers for storage developers who must navigate different rule sets across regions, making it difficult to accurately model forward revenue projections, develop consistent operating strategies, or achieve economies of scale in fleet management.

The Commission should direct RTOs/ISOs to eliminate artificial operational limitations and instead focus on three principles:

1. Allow BESS resources to operate across their full technical envelope as defined by manufacturer specifications and resource adequacy commitments;
2. Implement performance-based mechanisms that hold resources accountable for delivery rather than imposing operational restrictions; and,
3. Provide clear transparency regarding system needs – both in day-ahead and forward markets – with clear pathways for resources to demonstrate their ability to meet those needs.

Award restriction is preferred over SOC management. Some RTOs/ISOs have implemented systems that directly manage SOC for BESS resources participating in certain markets or products. While this approach may appear to provide operational certainty, it creates significant problems in practice and represents a fundamental departure from how other market resources are treated.

RTO/ISO SOC management typically occurs at the worst possible time from a financial standpoint, often forcing charging during high-price periods or discharging during low-price periods to maintain SOC targets that may or may not be necessary for reliability. These management actions are based on system forecasts and assumptions that may not align with individual resource characteristics, capabilities, or the operator's own forecasting and optimization. The result is forced economic losses for the resource owner while simultaneously reducing overall system efficiency.

This approach also creates moral hazard by socializing the consequences of poor operational decisions. When the RTO/ISO manages SOC, resources are effectively relieved of responsibility for their own operational optimization. This can lead to less efficient operators succeeding in the market while more sophisticated operators find their capabilities underutilized, ultimately harming the competitive development of the storage sector.

A more prudent approach is to hold resources accountable for their commitments through performance-based mechanisms and award restrictions. If a resource commits to provide a service, it should be responsible for managing its own SOC to ensure delivery, with clear consequences for non-performance. Award restrictions that prevent economically or physically infeasible commitments protect system reliability without unnecessarily intervening in resource operations.

This approach aligns with how conventional resources are treated. A gas generator is not guaranteed fuel delivery by the RTO/ISO, but it is held accountable for showing up when committed. The same principle should apply to BESS. The resource operator is best positioned to manage its "fuel" (state of charge), and market rules should trust operators to do so while maintaining clear penalties for failure to deliver.

The Commission should establish a clear principle that RTOs/ISOs should not directly manage SOC for BESS resources except in emergency conditions. Instead, market rules should focus on appropriate award restrictions, performance requirements, and penalties that hold resources accountable while allowing them to manage their operations to maximize both private value and system benefit.

New technologies should not be penalized for shortcomings in regulatory structure, operations, and/or modeling. The integration of energy storage into wholesale markets has followed a troubling pattern repeated across multiple RTOs/ISOs:

1. New technology emerges;
2. Existing market rules and systems are applied with minimal adaptation;
3. Resources operate under these loose frameworks;
4. Operational or market issues emerge;
5. The RTO/ISO attributes problems to the technology rather than the market design; and,
6. Artificial limitations are imposed that suppress the technology's development and deployment.

This pattern is particularly concerning for BESS because it is not an incremental technology evolution but, rather, it represents a fundamentally different resource type that directly addresses the grid's need for flexibility, fast response, zero-emission energy arbitrage, and the ability to provide multiple services simultaneously or in rapid succession. When market design inadequacies create operational problems, the response should not be to limit the technology but to fix the market design.

The current reactive approach imposes real costs on the energy transition. Artificial limitations imposed after projects are operational can destroy carefully underwritten business cases, leading to financial distress for early adopters and creating significant regulatory risk that deters future investment. This directly contradicts the Commission's objectives in Order 841 and subsequent proceedings to remove barriers to storage participation.

RTOs/ISOs must become more proactive and sophisticated in their integration planning for new technologies. This includes conducting thorough ex-ante analysis of how new resource types will interact with existing market rules, developing implementation plans that address known gaps before they create operational problems, maintaining flexibility to adapt quickly when unanticipated issues arise, and defaulting toward market-based solutions and performance requirements rather than administrative limitations when problems do emerge.

The Commission should require RTOs/ISOs to develop and submit technology integration frameworks that demonstrate how they will proactively adapt market rules and systems to accommodate energy storage and other emerging technologies. These frameworks should be subject to stakeholder review and should include commitments to avoid reflexive limitation of new technologies when market design inadequacies are the root cause of observed problems.

Transmission service charges for energy storage charging. MISO's practice of charging transmission service for energy storage withdrawals represents a significant barrier to storage development and is potentially in conflict with Order 841's (specifically 841a) directive regarding comparable treatment of energy storage.

When energy storage withdraws energy from the grid, it is providing a valuable service to the system. This withdrawal may be smoothing net demand, relieving transmission congestion by reducing demand in constrained areas, providing downward energy or regulation service in response to RTO/ISO dispatch, or positioning energy for later dispatch when the system needs it. This is fundamentally different from serving firm load.

Energy storage charging is dispatchable load responding to market price signals and RTO/ISO dispatch instructions. It is not firm load that must be served at all times. The energy withdrawn in one interval may be converted directly into an upward ancillary service in the next interval, providing critical flexibility that enhances grid reliability. Charging transmission service for these withdrawals creates a substantial economic penalty that discourages exactly the kind of flexible, grid-responsive operation that storage is uniquely positioned to provide.

Order 841 explicitly addresses this issue, stating that when an energy storage resource is providing a service to the market and is dispatched to withdraw energy, it should not be charged transmission service. While the Commission approved MISO's initial compliance filing, the fundamental tension with Order 841's principles remains.

No other RTO/ISO applies transmission charges to storage charging in the manner MISO does, creating a significant competitive disadvantage for MISO and a barrier to IPP-based storage development in the region. This is particularly problematic as MISO faces significant reliability challenges that storage could help address, and the region is attempting to integrate large amounts of renewable generation that would benefit from co-located or regional storage deployment.

The Commission should revisit this issue and provide clear guidance that energy storage resources participating in RTO/ISO markets and responding to dispatch instructions should not be assessed transmission charges for withdrawals, consistent with the principles articulated in Order 841.

ADDITIONAL COMMENTS ON THE VALUE OF BATTERY STORAGE TO SUPPORT LARGE LOAD INTERCONNECTION

Reliability. BESS is a reliable capacity resource that can serve loads with far greater flexibility and response speeds than traditional capacity resources. Energy storage projects can ramp up from 0% to 100% output in milliseconds, have forced outage rates below 1%, are not impacted by fuel logistics or price volatility, and can be sited wherever they are needed. Each of these attributes align with needs posed by data centers that require power supply at higher reliability standards than typical utility customers. Grid planners and operators must ensure these needs are met through capacity resources like BESS.

BESS also complements traditional generating resources to ensure system reliability during peak conditions, when energy demand surges, or when supply is reduced due to outages. Storage also makes every generating resource – thermal or renewable – more efficient and, therefore, more affordable. Storage makes thermal plants more efficient by smoothing their output and avoiding frequent start-ups and shutdowns, while it can also firm output from renewable generation.

Affordability. Battery energy storage enhances affordability by firming intermittent renewable generation, maximizing the efficient use of baseload power, and reducing the need for expensive transmission upgrades, ultimately delivering reliable and cost-effective electricity for large-scale consumers.

- **Firming Intermittent Generation.** BESS charges during off-peak hours, when cheap renewable energy is abundant, and discharges during peak demand periods when electricity is at its most expensive. In addition to firming intermittent renewable generation, BESS deployment necessarily means that fewer peaking plant hours are

required, which in turn reduces peaking plant maintenance costs and capital expenditures.
- **Extending Use of Baseload Power**. Frequent cycling of gas plants leads to increased wear and tear, higher maintenance costs, and reduced operational efficiency. BESS mitigates these issues by absorbing excess generation, allowing gas plants to run steadily and minimizing the need for costly and inefficient start-ups and shutdowns.
- **Mitigating Need for Transmission Upgrades.** BESS can discharge during peak demand periods, shaving the net load seen by the upstream transmission network and mitigating the need for transmission upgrades;

Storage as a Transmission Asset. As noted above, the Commission considers BESS as equivalent to a generating asset within the context of its Large Generator Interconnection Procedures, and it should do the same within this large loads context to preempt the need to create a new regulatory procedures specific to storage. Additionally, it is critical for the Commission to recognize the attributes of storage when deployed as a transmission asset.

Large loads, including those associated with data centers, can face multi-year interconnection delays driven by transmission-related constraints. When strategically deployed, BESS can provide a faster, more flexible, and lower cost alternative to traditional transmission upgrades.

- BESS located near data center loads can discharge during peak demand periods, shaving the net load seen by the upstream transmission network and mitigating the need for transmission upgrades;
- BESS can absorb excess power during high-generation periods, lowering real-time flows on a constrained path;
- BESS can serve as a fast-response asset capable of absorbing instantaneous oversupply when large loads trip offline, which would otherwise cause voltage disturbances;

BESS as transmission can help to support data centers that increase load when built out over multiple phases, allowing that load growth to occur much faster than new transmission line buildout.

In short, BESS can accelerate large load interconnection by solving many of the transmission constraints that are currently causing interconnection delays. Strategically sited storage can enable large loads to interconnect years earlier while fully maintaining system reliability.

Encourage storage as more than a transmission-only asset. Most RTOs/ISOs that have addressed energy storage as a transmission asset have imposed restrictions requiring these resources to participate exclusively as transmission assets, commonly referred to as Storage as Transmission-Only Assets (SATOA) in some markets. While we recognize there are legitimate considerations in designing a framework for dual-use storage, the current transmission-only participation model significantly undervalues these resources and creates barriers to efficient grid solutions.

Energy storage is particularly well-suited to solving economic congestion relief and load pocket supply issues. These transmission challenges share several critical characteristics that make them ideal for storage solutions.

1. They are predictable. The conditions that trigger these constraints are known and can be forecasted with reasonable accuracy based on load patterns, generation dispatch, and weather conditions.
2. They are controllable. The solution involves generation movement in a specific direction (injection or withdrawal) at specific locations.
3. They are temporal. These constraints typically occur during specific hours of the day or seasons of the year, not continuously.

The temporal nature of transmission constraints is crucial to understanding the inefficiency of transmission-only participation models. A storage facility sized to address a transmission constraint may only be needed for that purpose during a few hundred hours per year, or in some cases only a few dozen hours. For a load pocket issue that manifests during summer peak periods, the SATOA may be critical for 3-4 hours per day for 60-90 days per year, totaling perhaps 300 hours annually. For the remaining 8,460 hours of the year, that asset sits idle under current rules, despite being capable of providing valuable energy, ancillary services, and ramping products to the grid.

Because transmission needs are forecastable and the grid operator can pre-plan for these events, there is no operational reason why a storage asset cannot participate in wholesale markets during non-constraint periods while maintaining the capability to perform its transmission function when needed. The RTO/ISO can establish clear requirements for state-of-charge management and availability during anticipated constraint periods, while allowing the resource to optimize its operations during all other hours. This is directly analogous to how conventional generation providing transmission support services (such as RMR units) can participate in energy markets when not needed for reliability.

The current transmission-only restriction creates several problems that harm both economic efficiency and grid reliability:

1. **Stranded Capacity**. Hundreds or thousands of megawatt-hours of energy storage capacity sit unused during non-constraint periods when they could be providing

critical flexibility to address renewable integration, provide frequency regulation, offer ramping capability, and support voltage control. At a time when system operators consistently emphasize the need for flexible resources, artificially sterilizing storage capacity makes no policy sense.

2. **Suppressed Investment**. The transmission-only participation model makes storage solutions to transmission problems more expensive than they need to be. If a developer can only recover costs through transmission rates or contracted capacity payments for transmission service, the project economics require higher payments than if the resource could earn additional revenue from wholesale market participation. This either makes storage solutions appear less competitive relative to traditional transmission upgrades, or requires ratepayers to pay more than necessary for the same transmission solution.
3. **Suboptimal Sizing**. Current rules create an incentive to build storage facilities sized exactly to the transmission need and no larger. However, a more efficient outcome might involve building a larger facility where a portion serves the transmission function while the remainder provides wholesale market services. For example, if 100 MW of storage can solve a transmission constraint, it might be economically optimal to build a 200 MW facility at the same location, reserving 100 MW (plus necessary energy capacity) for the transmission function while allowing the remaining 100 MW to participate fully in wholesale markets. The incremental cost of additional capacity at an existing site is often relatively low, but current rules provide no pathway for this hybrid model.
4. **Barrier to Transmission Solutions**. By making storage-based transmission solutions less economic than they could be, the current framework may cause system planners to default to traditional transmission upgrades even when storage would provide a lower-cost or more flexible solution. This is particularly problematic for transmission constraints that may be temporary (such as those that will be resolved by planned generation retirements or transmission upgrades in future years), where a storage solution could provide a lower-cost bridge than accelerating permanent transmission infrastructure.

A superior framework would allow energy storage serving transmission functions to participate in wholesale markets subject to appropriate restrictions that ensure transmission reliability is not compromised. Key elements of such a framework would include:

- **Availability Requirements**. The storage resource must maintain sufficient state of charge and availability to meet anticipated transmission needs, with requirements established based on historical constraint patterns and forecasted system conditions.

- **Dispatch Priority**. When the transmission constraint is binding or forecasted to bind within a defined look-ahead period, the transmission function takes priority over wholesale market participation, with the resource obligated to maintain the required state of charge and respond to transmission-related dispatch instructions.
- **Performance Accountability**. Clear consequences for failure to perform the transmission function when needed, with penalties sufficient to ensure the resource maintains appropriate availability.
- **Market Participation Parameters**. Outside of constraint periods and look-ahead windows, the resource can submit bids and offers for energy and ancillary services consistent with its available capacity after reserving what is needed for transmission obligations.
- **Transparent Forecasting**. The RTO/ISO provides clear information about anticipated constraint periods, allowing the resource operator to optimize market participation while maintaining transmission availability.

This approach is consistent with how other dual-function resources are treated in wholesale markets and would unlock significant value from storage investments without compromising transmission reliability. It would also better align with Order 841's directive to remove barriers to storage participation and ensure that storage resources can provide all services they are technically capable of providing.

The Commission should direct RTOs/ISOs to allow transmission-only participation requirements for energy storage serving transmission functions and, additionally, develop frameworks that allow these resources to provide wholesale market services when not needed for transmission purposes, subject to appropriate availability and performance requirements.

In short, unless the Commission directs RTOs/ISOs to allow for energy storage resources to provide both whole market and transmission services, we will be limited in how much value we are able to derive from storage as a transmission asset.

ELECTRIC GRID INFRASTRUCTURE SUPPORTING LARGE LOADS MUST BE DESIGNED TO MAXIMIZE GRID SECURITY

The security of electric grid infrastructure – and especially the portions of that infrastructure that support large loads like data centers and manufacturing – are critical to national security. As the grid becomes more interconnected, the Commission should consider taking action to address the threat posed by global proliferation of potentially compromised controls hardware and software that is designed or manufactured by entities connected with foreign adversaries. This is especially true for BESS, which is highly digital, interconnected, and relies on software-based controls.

BESS introduces significant advantages — enabling faster response times, more efficient system management, and better integration of new technologies – as described above. It is imperative that these assets are secure and protected from cyberattacks as they become a more critical component of the nation's power infrastructure. Sabotage of critical components in a single asset via pre-installation or remote deployment of malicious firmware, for example, could have cascading effects on grid reliability. Systems can and should be designed from the outset to maximize security by eliminating these upstream vulnerabilities in the supply chain.[4]

Cybersecurity should be embedded into every stage of project development and procurement given the sensitivity of large electric loads. Fluence encourages FERC to take action to ensure that supply chains supporting the infrastructure powering large loads does not include hardware and software controls from foreign adversaries.

CONCLUSION

BESS must play a critical role if we are to ensure the reliable and efficient interconnection of large loads. To maximize the value of energy storage, regulatory frameworks must be created that recognize the unique operational aspects of BESS, promote market designs that fully value is flexibility, and move barriers to its deployment as both a generation asset and a transmission asset. Fluence encourages the Commission to create expedited interconnection pathways for hybrid facilities, eliminate artificial operational and market limitations, and adopt policies that enable dual-use storage to maximize system benefits. Commission action can unlock the full potential of energy storage to meet the evolving demands of the electric grid and large load customers.

[4] Hutton, Katherine, and Weiner, Michael. *Cybersecurity in Battery Energy Storage: Mitigating Risks in a Changing Policy Landscape.* Fluence, 2025. https://blog.fluenceenergy.com/cybersecurity-battery-energy-storage-mitigating-risks-changing-policy-landscape.